

From fossil pollen to a picture of past vegetation

Project progress and decisions for our first classifier

Identify pollen types in microscope images.

Support research on North American vegetation about 50–60 million years ago.

Today: what we have → what we need to agree



Finding a grain and naming it are different tasks

PREVIOUS WORK

Find the specimen

Locate an object within a large microscope image.



THIS SEMESTER

Identify its type

Assign a useful category to an already located specimen.



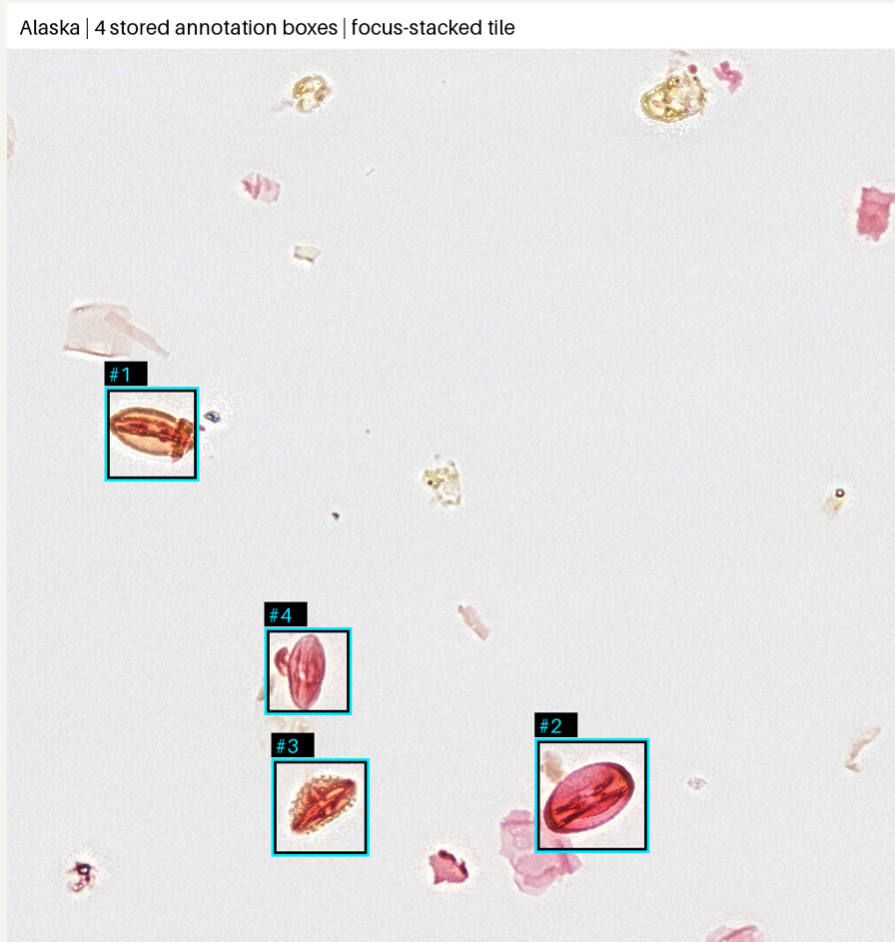
SCIENTIFIC USE

Review the result

Let an expert inspect the image, category and uncertain cases.

Does this capture the most useful focus for this semester?

We can view small regions of the large slides



The boxes tell us where to look.

These examples are labeled “pol” (broad pollen), not a detailed type.

Images stay on Rice’s cluster.

We read small regions instead of downloading whole slides.

The label inventory is ready

We connected slide files, specimen locations and names to our working categories.

128

Annotation files linked to
slide images

933

Annotations in our YES
working set

27

Priority categories
represented

Next: extract and inspect the specimen images.

The labeled image dataset is not yet prepared for training.

Inventory completed • example images viewed • no classifier trained yet

Some categories have much stronger coverage

Examples from our current working inventory

CATEGORY	ANNOTATIONS		SLIDES
Bisaccate		335	44
Momipites sp		262	40
Caryapollenites sp.		39	15
Monocolpites sp.		33	12
Rhoipites sp.		39	1

Which scarce categories are essential, and could more examples be labeled?

Turn the inventory into a training dataset

We will use the agreed working categories and keep the original labels for traceability.

01

Extract the images

Read a small region around each selected annotation, including marks outside the old training areas.

02

Check the examples

Inspect position, focus and image quality; look for duplicate or unusable specimens.

03

Prepare a fair test

Choose supported categories and reserve slides for evaluation before training.

Which visual features and preservation problems should we check first?

What is most important for us to get right?

01

Critical distinctions

Which pollen types must we be especially careful not to confuse?

02

Fair testing

Check performance on specimens from slides the model did not learn from.

03

Uncertain cases

Allow a specimen to be flagged for expert review when the prediction is uncertain.

For large categories, should ~60 specimens apply to training only?

A simple result you can inspect

INPUT

A marked specimen

An image region with a known location on the original slide.

RESULT

A proposed category

The predicted type, with uncertain cases clearly flagged.

REVIEW

Image + identification

A simple viewer and a new annotation file; originals preserved.

First version: classify existing marked specimens.

Which viewing software and export format would fit your workflow?

All 27 represented YES categories

Current working annotation counts

CATEGORY	COUNT	SLIDES	CATEGORY	COUNT	SLIDES
Bisaccate	335	44	Bombacacidites sp.	10	7
Momipites sp	262	40	Inaperturopollenites sp.	10	2
Caryapollenites sp.	39	15	Caryapollenites veripites	7	3
Rhoipites sp.	39	1	Laevigatosporites sp.	6	4
Monocolpites sp.	33	12	Illexpollenites sp.	5	4
Alnipollenites verus	31	11	Tetracolporopollenites sp.	3	1
Cupuliferoipollenites sp.	29	1	Ulmipollenites sp.	2	1
Psilatriletes sp.	20	13	Momipites wyomingensis	1	1
Cicatricosisporites sp.	19	6	Pistillipollenites sp.	1	1
Ericipites sp.	18	7	Platycaryapollenites swasticooidus	1	1
Quercus sp.	18	1	Polyatriopollenites type	1	1
Betulaceae	17	12	Retipollenites sp.	1	1
Arecipites sp.	13	9	Taxodium sp.	1	1
Platycarya platycarioides	11	5			

The main questions to leave with

Agree on priorities, next actions and who can help.

- 1 What matters most?**
Which pollen types must we get right, and which mistakes matter most?
- 2 Where do we need more examples?**
Which scarce categories are essential? Could more specimens be labeled?
- 3 What should we check in the images?**
Which visual features and preservation problems should guide our checks?
- 4 How should we start and test fairly?**
Start with better-supported categories? Apply the ~60 limit to training only?
- 5 What would be most useful to receive?**
Which viewer, annotation format and review of uncertain cases fit your work?

Before we finish: confirm an owner and next step for each open question.